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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Assume that $z = f(w)$, $w = g(x, y)$, $x = 3r^3 - s^2$, and $y = re^s$. If $g_x(3, 1) = -2$, $g_y(3, 1) = 7$, $f'(8) = -4$, and $g(3, 1) = 8$, find the following.

$$\left. \frac{\partial z}{\partial r} \right|_{r=1, s=0} \quad \text{and} \quad \left. \frac{\partial z}{\partial s} \right|_{r=1, s=0}$$

Suppose that $w = g(x, y)$, $x = h(r, s)$, and $y = k(r, s)$. If all three functions are differentiable, then w has partial derivatives with respect to r and s , as given by the following formulas.

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \quad \text{and} \quad \frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s}$$

The expression $\left. \frac{\partial z}{\partial r} \right|_{r=1, s=0}$ represents the derivative of z with respect to r , evaluated at $r = 1$, $s = 0$.

The derivative $\frac{\partial z}{\partial r}$ can be rewritten as $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial w} \frac{\partial w}{\partial r}$.

Substituting $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$ into $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial w} \frac{\partial w}{\partial r}$, the derivative $\frac{\partial z}{\partial r}$ can be rewritten as $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial w} \left(\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \right)$.

The derivative f' in $f'(8) = -4$ can be rewritten as $f' = \frac{\partial z}{\partial w}$.

The derivative g_x in $g_x(3, 1) = -2$ can be rewritten as $g_x = \frac{\partial w}{\partial x}$.

Similarly, the derivative g_y in $g_y(3, 1) = 7$ can be rewritten as $g_y = \frac{\partial w}{\partial y}$.

Given that $x = 3r^3 - s^2$, find $\frac{\partial x}{\partial r}$.

$$\frac{\partial x}{\partial r} = 9r^2$$

Given that $y = re^s$, find $\frac{\partial y}{\partial r}$.

$$\frac{\partial y}{\partial r} = e^s$$

When $r = 1$ and $s = 0$ the value of x is 3.

When $r = 1$ and $s = 0$ the value of y is 1.

Therefore, $r = 1$ and $s = 0$ corresponds to the point $(3, 1)$.

When $r = 1$, $s = 0$, and $(x, y) = (3, 1)$, the value of w is $g(3, 1) = 8$.

When $r = 1$, $s = 0$, $(x, y) = (3, 1)$, and $g(3, 1) = 8$, the value of f' is -4 .

Substitute each of the calculated terms into the equation and evaluate at the given point.

$$\begin{aligned} \left. \frac{\partial z}{\partial r} \right|_{r=1, s=0} &= \left. \frac{\partial z}{\partial w} \left[\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \right] \right|_{r=1, s=0} \\ &= f' \left[(g_x) (9r^2) + (g_y) (e^s) \right] \Big|_{r=1, s=0} \\ &= (-4) [(-2)(9(1^2)) + (7)(e^0)] \\ &= 44 \end{aligned}$$

Solve for $\frac{\partial z}{\partial s} \Big|_{r=1, s=0}$ using a similar process. Start by rewriting $\frac{\partial z}{\partial s}$ as $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial w} \left(\frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \right)$. Note that $\frac{\partial z}{\partial w}$, $\frac{\partial w}{\partial x}$, and $\frac{\partial w}{\partial y}$ also appear in the derivative of $\frac{\partial z}{\partial r}$.

Given that $x = 3r^3 - s^2$, find $\frac{\partial x}{\partial s}$.

$$\frac{\partial x}{\partial s} = -2s$$

Given that $y = re^s$, find $\frac{\partial y}{\partial s}$.

$$\frac{\partial y}{\partial s} = re^s$$

Substitute each of the calculated terms into the equation and evaluate at the given point.

$$\begin{aligned} \frac{\partial z}{\partial s} \Big|_{r=1, s=0} &= \frac{\partial z}{\partial w} \left[\frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \right] \Big|_{r=1, s=0} \\ &= f' \left[(g_x)(-2s) + (g_y)(re^s) \right] \Big|_{r=1, s=0} \\ &= (-4) \left[(-2)(-2 \cdot 0) + (7)(1 \cdot e^0) \right] \\ &= -28 \end{aligned}$$

Therefore, $\frac{\partial z}{\partial r} \Big|_{r=1, s=0} = 44$ and $\frac{\partial z}{\partial s} \Big|_{r=1, s=0} = -28$.